EE 132 — Automatic Control

Homework 6 : Lead & Lag Compensator Design

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All three problems use the unity-feedback configuration of Fig. 1, with plant $G(s)$ and series compensator $D(s)$. Designs are carried out on the root locus (angle and magnitude conditions) and verified numerically; equivalent MATLAB code is listed with each solution.

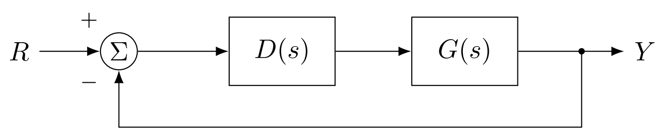

Figure 1: Unity-feedback configuration (Fig. 5.60).

Problem 26. Numerically controlled machine-tool servomechanism

The normalized plant is

$$G(s) = \frac{1}{s(s+1)},$$

and the closed-loop poles are to be placed at $s = -1 \pm j\sqrt{3}$ (i.e. $\omega_n = 2$, $\zeta = \frac{1}{2}$).

Solution

(a) Proportional control is insufficient. With $D(s) = k_p$ the loop gain is $L(s) = \dfrac{k_p}{s(s+1)}$ and the characteristic equation is

$$s(s+1) + k_p = 0 \implies s^2 + s + k_p = 0.$$

By Vieta's formula the two roots always sum to -1, so a complex-conjugate pair has

$$\mathrm{Re}(s) = -\tfrac{1}{2} \quad \text{for every } k_p.$$

The target poles require $\mathrm{Re}(s) = -1$, which is unreachable. Equivalently, on the root locus the angle condition at $s_0 = -1 + j\sqrt{3}$ gives

$$\angle G(s_0) = -\angle s_0 - \angle(s_0 + 1) = -120^\circ - 90^\circ = -210^\circ \equiv 150^\circ \neq \pm 180^\circ,$$

so s_0 is *not* on the locus. The angle deficiency is $\phi = 180^\circ - 150^\circ = 30^\circ$, which a lead network must supply.

(b) Lead design $D(s) = K\dfrac{s+z}{s+p}$. Cancel the plant pole at -1 by placing the zero at $z = 1$. The zero then contributes $\angle(s_0 + 1) = +90^\circ$, so the pole must supply the remaining $90^\circ - 60^\circ = 30^\circ$ net, i.e. $\angle(s_0 + p) = 60^\circ$:

$$\tan 60^\circ = \frac{\sqrt{3}}{p-1} = \sqrt{3} \implies p = 2.$$

The magnitude condition $|L(s_0)| = 1$ with $L(s) = \dfrac{K}{s(s+2)}$ gives

$$\frac{K}{|s_0|\,|s_0+2|} = \frac{K}{(2)(2)} = 1 \implies K = 4.$$

Hence

$$\boxed{D(s) = \frac{4(s+1)}{s+2}}, \qquad L(s) = \frac{4}{s(s+2)}, \qquad s^2 + 2s + 4 = 0 \;\Rightarrow\; s = -1 \pm j\sqrt{3}.$$

The poles land *exactly* on the specification. The velocity constant is $K_v = \lim_{s\to 0} sL(s) = 4/2 = 2$, giving a ramp error of 0.5. Numerical check: $\zeta = 0.5$, $M_p = 16.3\%$ (Fig. 3, 2).

Listing 1: Problem 26 — MATLAB

```
s = tf('s');
G = 1/(s*(s+1));
% (b) lead: zero cancels pole at -1, pole at -2, gain K=4
D = 4*(s+1)/(s+2);
L = D*G;  T = feedback(L,1);
damp(T)              % check poles at -1 +/- j*sqrt(3)
Kv = dcgain(s*L)     % velocity constant = 2
figure, rlocus(L),  sgrid(0.5,2)
figure, step(T)
```

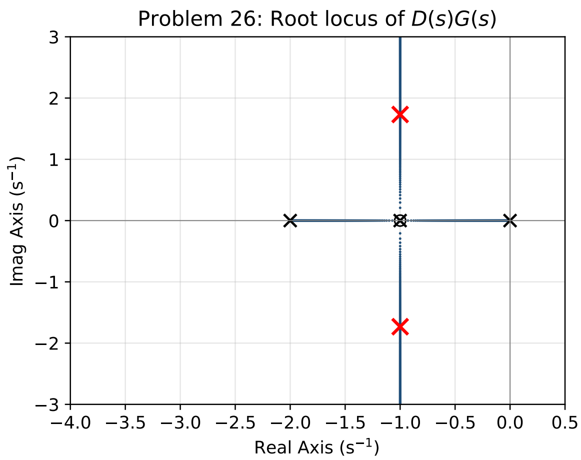

Figure 2: P26 root locus of $D(s)G(s)$; $\times$ marks the design poles.

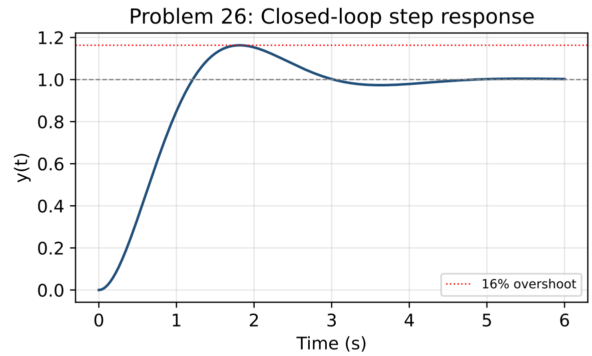

Figure 3: P26 closed-loop step response.

Problem 27. Servomechanism position control — lead + lag

$$G(s) = \frac{10}{s(s+1)(s+10)}.$$

Closed-loop specifications:

- step overshoot $M_p \le 16\% \;\Rightarrow\; \zeta \ge 0.5$;
- rise time $t_r \le 0.4$ s $\;\Rightarrow\; \omega_n \ge 1.8/t_r = 4.5$ rad/s;
- ramp error $e_{ss} < 0.02 \;\Rightarrow\; K_v > 50$.

Solution

Design target. Because the lead zero raises the actual overshoot above the second-order estimate, I aim inside the boundary: $\zeta = 0.65$, $\omega_n = 5.5$, which keeps M_p comfortably under 16% while giving ω_n enough for the rise-time spec:

$$s_0 = -\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} = -3.58 + j4.18.$$

(a) Lead compensation. Cancel the plant pole at -1 with the lead zero ($z = 1$). The open loop becomes $L(s) = \dfrac{10K}{s(s+10)(s+p)}$. With $\angle s_0 = 130.5°$ and $\angle(s_0 + 10) = 33.0°$, the angle condition needs

$$\angle(s_0 + p) = 180° - 130.5° - 33.0° = 16.4° \implies p = -\mathrm{Re}(s_0) + \frac{\mathrm{Im}(s_0)}{\tan 16.4°} = 17.76.$$

The magnitude condition $|L(s_0)| = 1$, i.e. $K = \dfrac{|s_0|\,|s_0+10|\,|s_0+p|}{10}$, gives $K = 62.36$, so

$$\boxed{D_{\text{lead}}(s) = \frac{62.36\,(s+1)}{s+17.76}}.$$

The lead-only closed-loop poles are $-3.55 \pm j4.15$ and -20.6; the transient specs are met ($M_p \approx 7\%$, $t_r \approx 0.38$ s), but $K_v = 62.36 \cdot \frac{1}{17.76} = 3.5$, far short of 50.

(b) Proportional control — velocity constant. For $D(s) = k_p$,

$$K_v = \lim_{s\to 0} s\,k_p\,G(s) = k_p \cdot \frac{10}{(1)(10)} = k_p.$$

Meeting $K_v > 50$ would require $k_p > 50$. But $k_p = 50$ gives the characteristic polynomial $s^3 + 11s^2 + 10s + 500$, whose Routh array has the s^1 entry $\frac{11\cdot 10 - 500}{11} = -35.5 < 0$: *two right-half-plane poles* ($s = 0.48 \pm j6.84$), i.e. the loop is unstable. Proportional control therefore cannot meet the steady-state requirement while remaining stable — dynamic (lead + lag) compensation is required.

(c) Lag compensation. The lag must raise K_v from 3.5 to above 50, a factor of at least 14; take $\alpha = z/p = 25$ for margin and place the dipole near the origin so the dominant transient is unchanged: $z = 0.05$, $p = z/\alpha = 0.002$. The complete compensator is

$$\boxed{D(s) = 62.36\,\frac{(s+1)(s+0.05)}{(s+17.76)(s+0.002)}}.$$

Then $K_v = 3.5 \times 25 \approx 88$, so $e_{ss} = 1/K_v = 0.011 < 0.02$. Numerical verification of the final design:

$$M_p = 7.4\%, \quad t_r = 0.38\text{ s}, \quad t_s \approx 1.2\text{ s}, \quad K_v = 88, \quad e_{ss} = 0.011,$$

with dominant poles $-3.55 \pm j4.15$, a fast pole at -20.6, and the lag pole/zero pair near -0.05. All three specifications are satisfied. The root locus and step response of the final design are shown in Fig. 4–5.

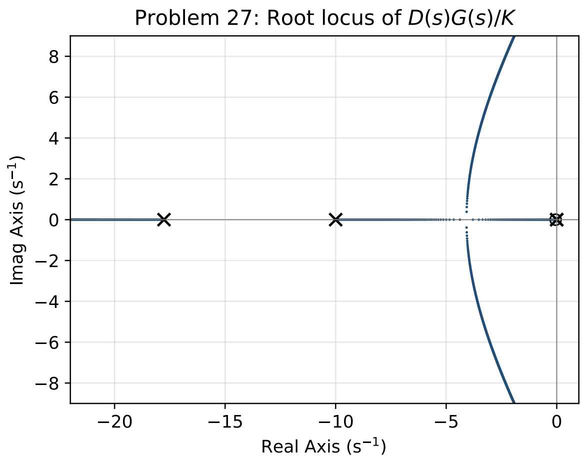

Figure 4: (d) P27 root locus of the final $D(s)G(s)$.

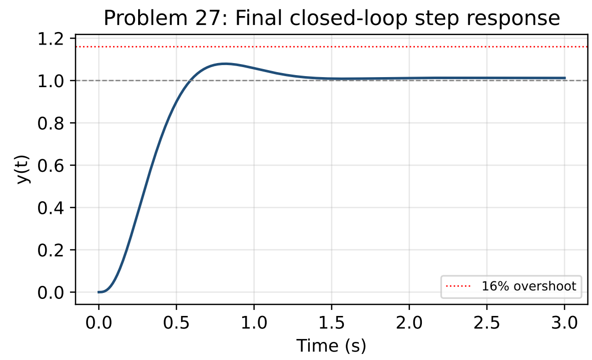

Figure 5: (e) P27 step response of the final design.

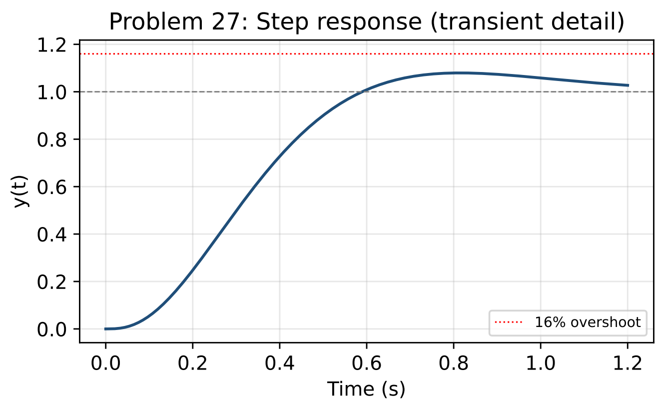

Figure 6: P27 step response, transient detail: $M_p = 7.4\%$ ($< 16\%$), $t_r = 0.38$ s (< 0.4 s).

Listing 2: Problem 27 — MATLAB

```
s = tf('s');
G = 10/(s*(s+1)*(s+10));

% --- (a) lead: zeta=0.7, wn=4.5, cancel pole at -1 ---
zeta = 0.65;  wn = 5.5;
s0 = -zeta*wn + 1j*wn*sqrt(1-zeta^2);
ang  = @(z) angle(z)*180/pi;
need = 180 - ang(s0) - ang(s0+10);          % required angle of (s0+p)
p = -real(s0) + imag(s0)/tand(need);         % p = 17.76
K = abs(s0*(s0+p)*(s0+10))/10;               % K = 62.36
Dlead = K*(s+1)/(s+p);

% --- (b) proportional Kv ---  Kv = kp ; kp=50 -> unstable
roots([1 11 10 500])                         % check RHP poles

% --- (c) lag: raise Kv past 50 ---
alpha = 25;  z = 0.05;  plag = z/alpha;
Dlag = (s+z)/(s+plag);
D = Dlead*Dlag;
L = D*G;  T = feedback(L,1);
Kv = dcgain(s*L)                             % ~88  -> ess = 0.011
stepinfo(T)                                  % Mp, tr, ts
figure, rlocus(L)                            % (d)
figure, step(T)                              % (e)
```

Problem 28. Lag design for prescribed dominant poles

$$G(s) = \frac{1}{s(s+2)},$$

with dominant closed-loop poles at $s = -1 \pm j$ and ramp error $e_{ss} < 0.2$.

Solution

Gain to meet the transient. With proportional gain K, $L(s) = \dfrac{K}{s(s+2)}$ and $s^2+2s+K=0$ gives $s = -1 \pm \sqrt{1-K}$. The real part is fixed at -1 for every K (the sum of roots is -2), so $-1 \pm j$ lies *exactly* on the locus and is obtained with

$$1 - K = -1 \implies K = 2 \qquad (\zeta = \tfrac{1}{\sqrt{2}} = 0.707,\ \omega_n = \sqrt{2}).$$

No lead is needed. The resulting velocity constant is $K_v = \lim_{s\to 0} sL(s) = K/2 = 1$, so $e_{ss} = 1$ — far too large.

Lag compensation. Add $D_{\text{lag}}(s) = \dfrac{s+z}{s+p}$ with ratio $\alpha = z/p$. Since $K_v = \frac{K}{2}\alpha = \alpha$, the requirement $e_{ss} < 0.2$ ($K_v > 5$) needs $\alpha > 5$; take $\alpha = 10$ for margin and place the dipole close to the origin so the dominant poles barely move: $z = 0.02$, $p = z/\alpha = 0.002$. The compensator is

$$\boxed{D(s) = \frac{2(s+0.02)}{s+0.002}}, \qquad K_v = 10, \qquad e_{ss} = \tfrac{1}{10} = 0.1 < 0.2.$$

The lag contributes only $\approx -1^\circ$ at $s_0 = -1+j$, so the dominant poles stay at $\approx -0.99 \pm j0.99$ ($\simeq -1 \pm j$); the third (lag) pole near -0.02 is almost cancelled by the zero at -0.02 and is non-dominant. The step and unit-ramp responses (Fig. 7–9) confirm the dominant pole placement and the reduced ramp error.

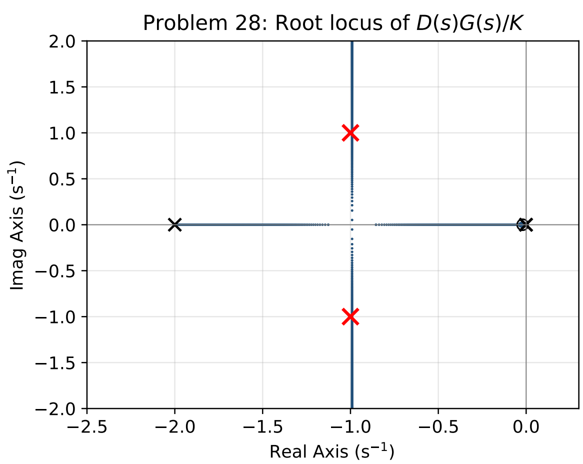

Figure 7: P28 root locus; $\times$ marks $-1 \pm j$.

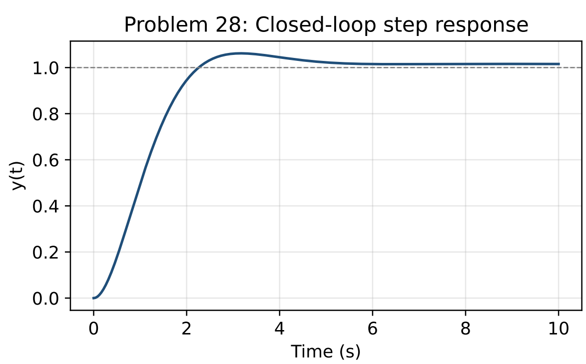

Figure 8: P28 closed-loop step response.

Listing 3: Problem 28 — MATLAB

```
s = tf('s');
G = 1/(s*(s+2));
K = 2;                          % places poles at -1 +/- j
% lag: alpha = 10 -> Kv = 10, ess = 0.1
z = 0.02;  p = 0.002;
```

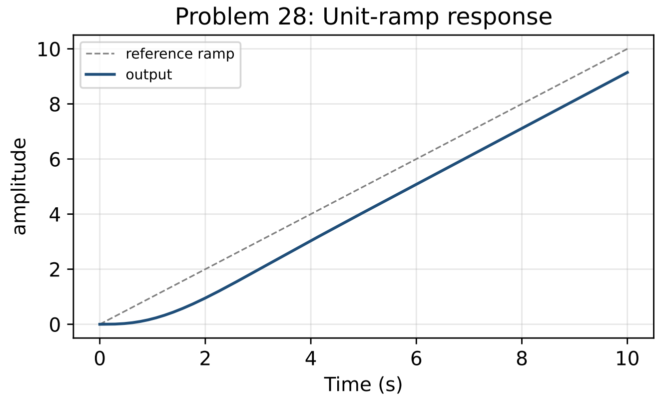

Figure 9: P28 unit-ramp response; the steady tracking error settles to $e_{ss} = 0.1$.

```
D = K*(s+z)/(s+p);
L = D*G;  T = feedback(L,1);
damp(T)                          % dominant poles ~ -1 +/- j
Kv = dcgain(s*L)                 % = 10  -> ess = 0.1
figure, rlocus(L), sgrid
figure, step(T)
t = 0:0.01:10;  figure, lsim(T,t,t)   % unit-ramp response
```